



Neurobotics

The Future of Cognitive Science

PIONEERING THE NEXT ERA OF BIOLOGICALLY-
INSPIRED DETERMINISTIC SYSTEMS

Neurobotics Founders



Kenneth Renshaw (CEO/CSO)

The mind behind Neurobotics. 50+ years of neuroscience, mathematics, and robotics study. Deep-tech engineering experience including work on the Galileo Space Probe, radiation-hardened ATAC16 deep-space computers, and key contributor to the TCAS II mid-air collision avoidance system design and deployment. Sun, Oracle, Yahoo, and SGI Silicon Valley engineer with over 40 years of modern systems engineering and DevOps expertise.



Shannon Renshaw (COO/CCO)

The creative heart of Neurobotics. Managing company operations and communications, as well as compliance. Author of the “Silicon Golden Rule”. Over 35+ years of institutional financial responsibility and administrative development experience ensuring integrity and ethical alignment of educational institutions, non-profits, emergency response, Silicon Valley start-ups, and established companies.



The Austere Vulnerability



The Problem

The current trajectory of computing relies on scaling cloud-tethered networks attached to massive data centers. These architectures face fundamental ceilings: they cannot guarantee outcomes, they cannot operate in extreme environments, and they cannot operate without connections to those massive data centers.

The Risk

Continuing to fund probabilistic "black box" models for autonomous applications guarantees a future of un-auditable, hackable, and fragile systems. Without a foundational shift to deterministic math in the hardware and software levels, next-generation kinetic assets will remain permanently tethered to vulnerable datalinks and data centers.



Monolithic Determinism



RISC-V/FPGA Substrate

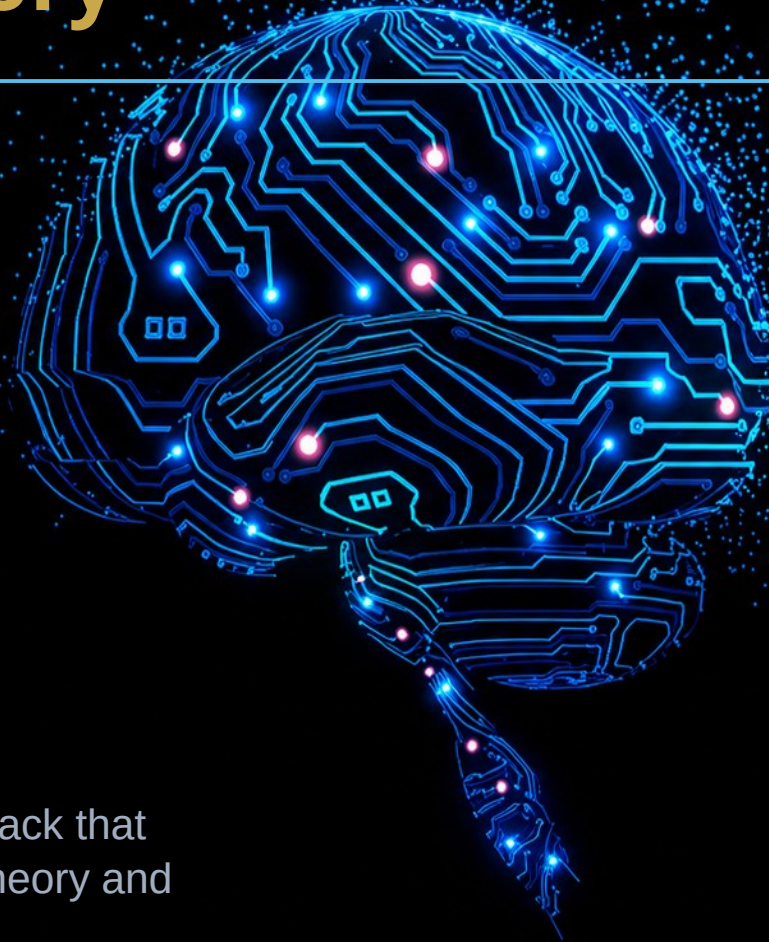
By eradicating probabilistic LLMs from the executive tier and processing 4K-D or 64K-D Vector Symbolic math natively within a hybrid RISC-V/FPGA architecture, we achieve a level of hardware-enforced mathematical certainty and near-zero latency that cloud-tethered models cannot reach. True autonomy requires true physics, not probability.

The Renshaw Theory



Achieving Deterministic Cognition

- Renshaw Theory extends Albus's NIST/NASA real-time control framework (RCS/CMAC) with hardware-native hyperdimensional computing inside a biologically implemented nervous system.
- SBUD and SBALF are two hardware configurations of the same cognitive stack that address both leading-edge cognition theory and rigorous determinism.



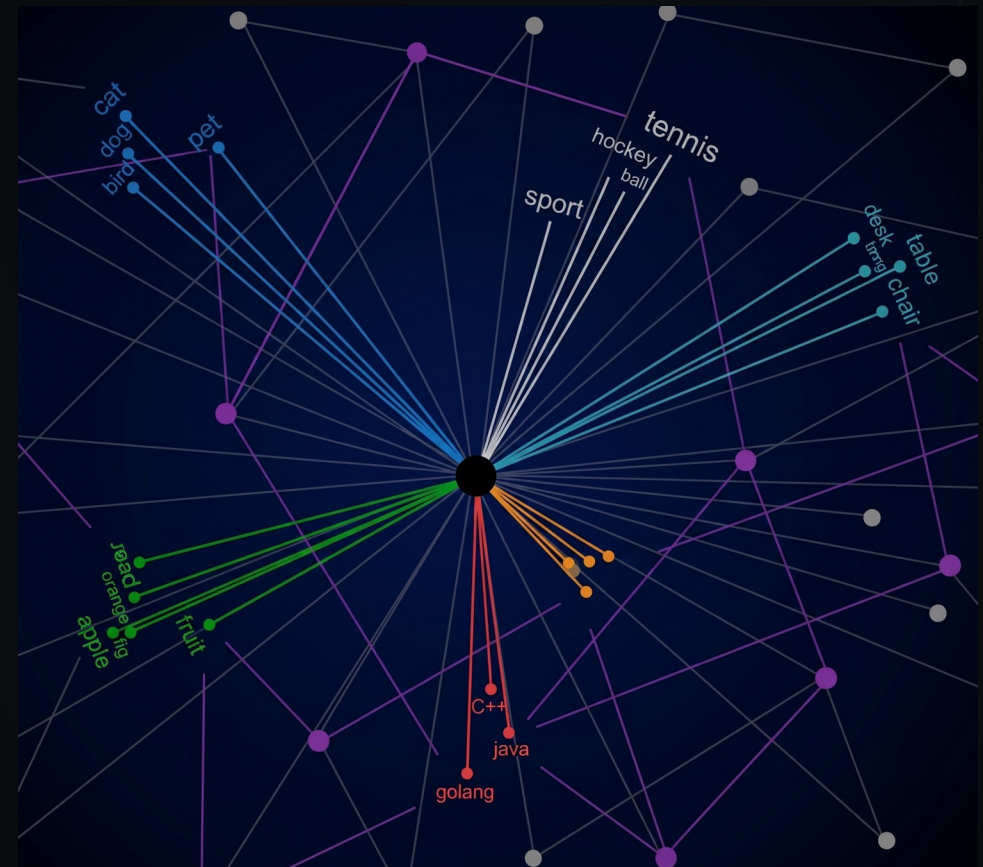
The SBUD Architecture



Edge-Native Sovereignty

In addition to our Silicon-Based Artificial Life Forms (SBALF) with it's complete cognitive stack, we offer the Silicon-Based Utility Droids (SBUD) for autonomous, high reliability operation in the harshest of environments. We build deterministic nervous systems, not black-box software.

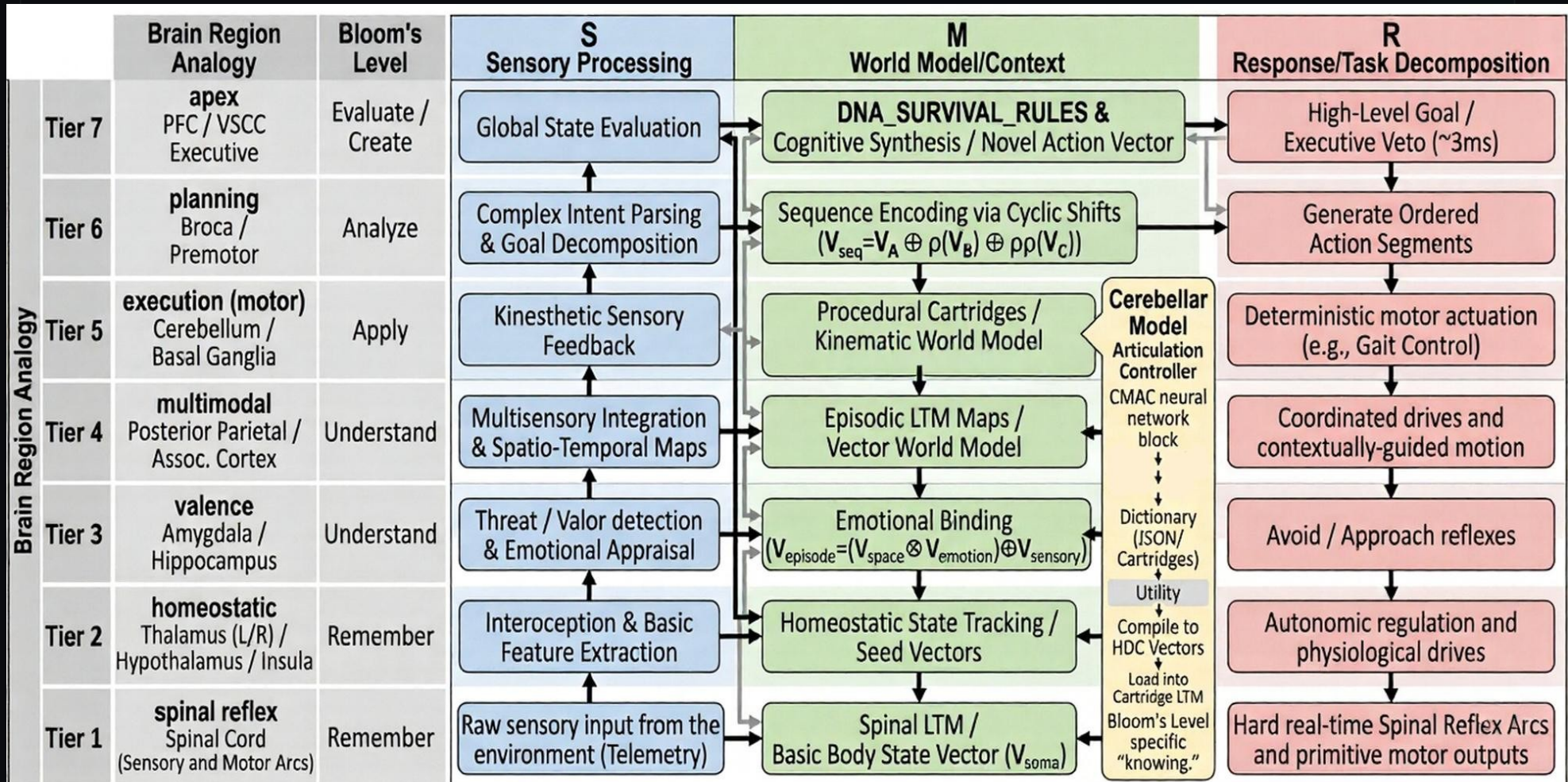
- **100% Off-Grid:** Operates locally with zero cloud reliance.
- **4K-D or 64K-D Vector Algebra:** Mathematical certainty mapping tactical Rules of Engagement directly to bare-metal.
- **Hardware Enforced Veto:** Executes natively in the pure RISC-V CPU pipeline with Tier 1 physical FPGA loop integration, providing a nearly instantaneous physical override immune to software hacking.



The Neuro-Cognitive Matrix



True Neuromorphic Biomimetic Cognition



7 Biological Tiers: From Spinal to Cognitive; Closely models human CNS; Real Neurotransmitter ODEs

7 Levels of Bloom's Taxonomy of Learning: Cognitive approach to creating internal world model; Overlays and complements Neuromorphic Tiers

7 Levels of Cross-Coupled, Holographic World Model-Driven Cognitive Control: Extension of and rooted in the work of James Albus at NSF and NIST Clearly and cleanly defined 7x3 matrix of cognitive modules.

The Market Reality



Unmanned Autonomous Vehicles

The global autonomous drone market is scaling aggressively, projected to reach \$163.6 Billion by 2030. The military drone sector alone is expected to reach \$109.2 Billion by 2031.



Humanoid Robotics

"They built the body, we built the brains." The global humanoid robot market is projected to hit \$38 Billion by 2035, with annual shipments exceeding 1.4 million units. Humanoids are the only scalable hardware interface to the \$60 Trillion global labor market.

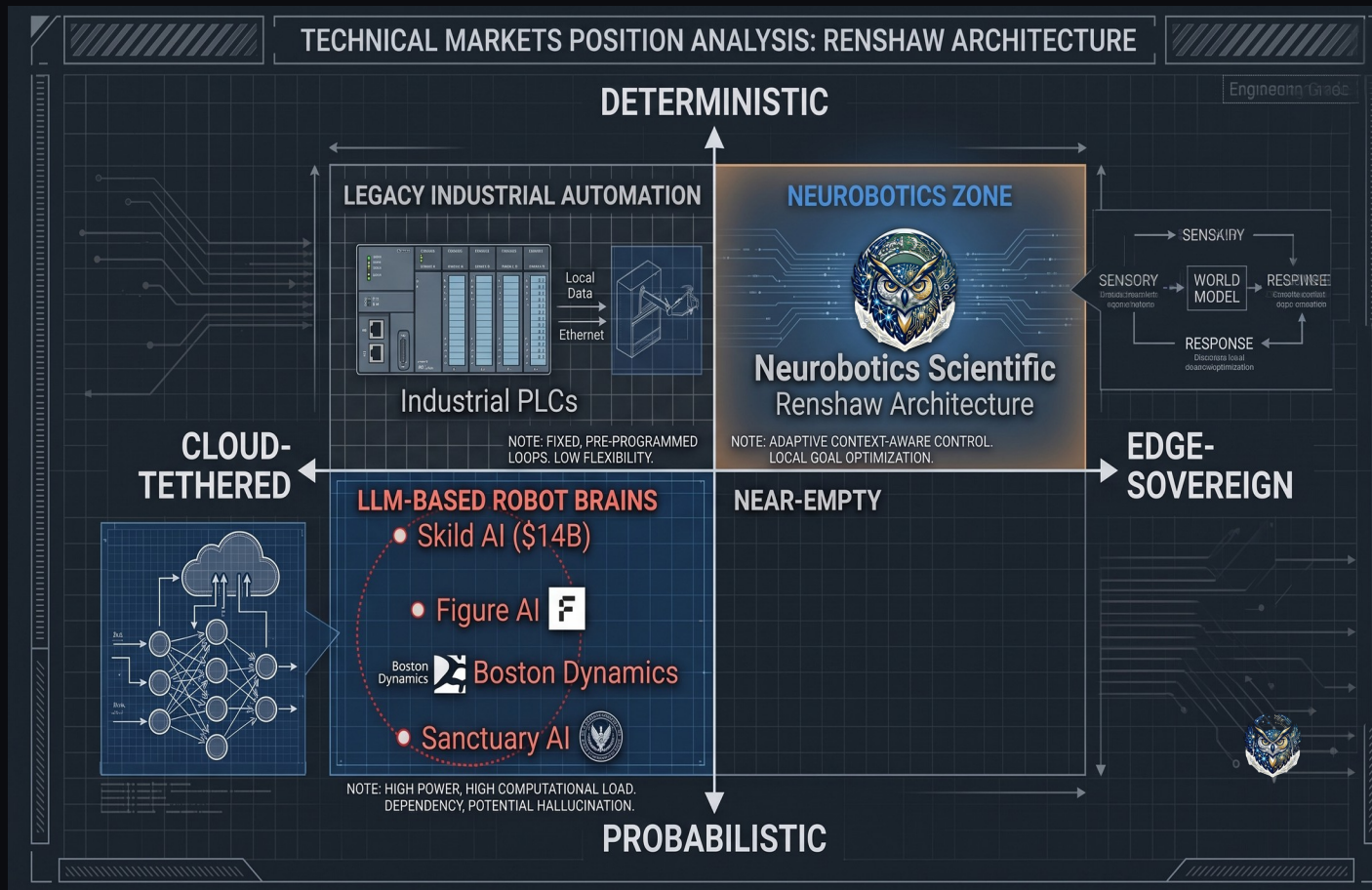


Energy & Industrial Inspection

The global inspection robotics market is projected to reach \$9.8 Billion by 2034, growing at nearly 14% annually. Specific sub-sectors like in-pipe and subterranean inspection are scaling toward \$19.2 Billion by 2035.



The Neurobotics Solution



Every competitor is building a smarter puppet. Neurobotics is building a nervous system.

Our competitors all share the same fundamental topology: a large probabilistic model (LLM or foundation model) living in a data center, communicating decisions down to an edge microcontroller via a network link.

The edge hardware executes, but it doesn't think. Cut the link and the robot freezes, hallucinates, or falls. The MCU at the edge is a relay, not a brain.

Skill's \$1.4B bet is on training a better model and shrinking the cloud closer to the robot — but the architecture is unchanged.

Neurobotics inverts this entirely. The cognitive hierarchy runs on the robot, from PRU spinal reflexes at 275µs through VSCC executive synthesis at the FPGA level, with zero datalink dependency. There is no "phone home".

The Modular Neurotronic Brain



Hardware-Enforced Determinism in a 100mm x 100mm Form Factor

Board One: The Somatic Controller (Tiers 1–5)

- **Function:** Drop-in lower nervous system for high-speed telemetry, sensor aggregation, and real-time spinal reflex arcs.
- **Key Feature:** Executes PRU spinal reflexes at 275 microseconds. Provides high-speed motor control without central processor bottlenecks.

Board Two: The SBUD Neocortex (Tiers 6–7)

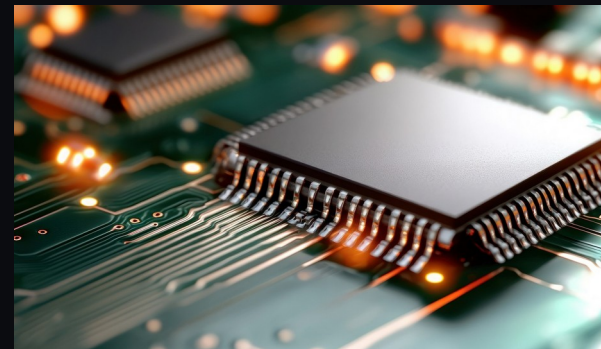
- **Function:** High-level executive synthesis, contextual reasoning, and mission-level task decomposition.
- **Key Feature:** Powered by a custom PIC64 / PolarFire SoC and our proprietary Aristotelian Syllogism Engine.

Boards Three & Four: Hardware Defense & Isolation

- **Galvanic Barrier (Board Four):** Optoisolated servo protection shield preventing inductive noise/spikes from reaching core logic.
- **SBALF Neocortex (Board Three):** Apex autopoietic cognitive engine. (*Strictly Confidential / Secured in Data Vault*).

The SBUD-SB2 Consolidation (Board Five):

Consolidates all 7 Tiers into a single 100mm x 100mm Single Board Brain. Operates fully *off-grid* under <25W peak power draw.



Commercial Dual-Licensing Framework



Balancing Open Science with Enterprise Monetization & Defense Security

Community Tier (AGPL-3.0)

- **Scope:** Free open-source drop-in for academic labs, researchers, and community testing (Board One).
- **Strategic Value:** Drives rapid academic adoption, hardware verification, and developer mindshare.

Enterprise & Defense Tier (MCLA)

- **Scope:** Master Commercial Licensing Agreements for enterprise robotics and defense integrators.
- **Strategic Value:** Direct revenue driver. Grants total copyleft exemption, allowing partners to integrate our Single Board Brain into closed-source, proprietary chassis.
- **Compliance:** Fully compliant with DISA-STIG, FIPS 140-3 cryptography, and zero-trust air-gapped standards.

Hardware Pilots (HEA)

- **Scope:** Hardware Evaluation Agreements for direct, hands-on enterprise proof-of-concept testing.
- **Strategic Value:** Rapid deployment of physical evaluation boards to field-test bare-metal SBUD units in hostile/off-grid environments.

Enterprise Licensing

Dual-Licensing Framework: Balancing
Open Science with Secure Commercial
Execution

The Neurobotics Moat:

High-margin IP licensing paired with hardware delivery creates a highly scalable revenue model with built-in copyleft defense.

The Neurobotics Pathway



Historical Execution & The Deterministic Roadmap to 2034

Phase I: The Foundation (Historical)

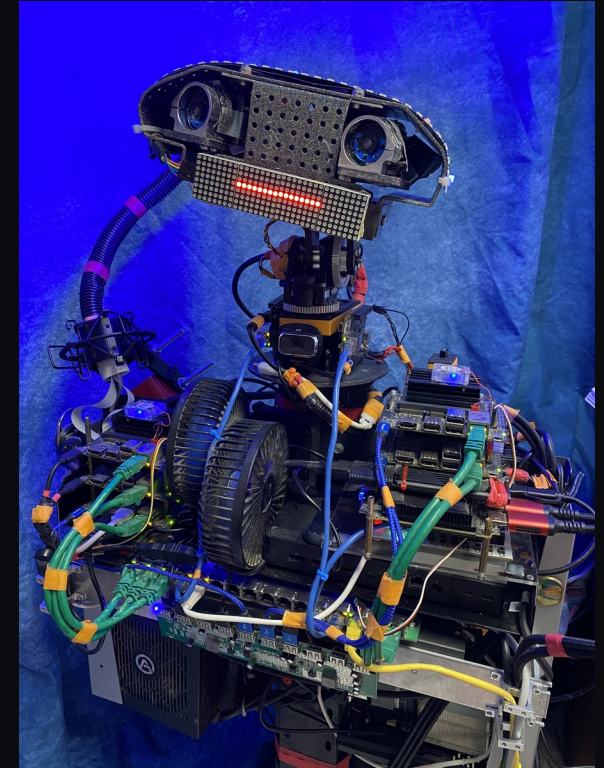
- **2019 Q3 | v1000:** *Bubo* Mark I Chassis Platform
- **2019 Q4 | v2000:** *Bubo* Mark VII Chassis Platform

Phase II: The Locus (Current & Seed Deployment)

- **2026 Q2–Q3 | v3000:** Neurotronic Brain Prototype Development
- **2026 Q2–Q3 | v4000:** *Bubo* Mark XX Chassis (*Convergence Locus*)
- **2026 Q3 | v5000:** Torso Humanoid VSA/HDC Platform (Without Arms)
- **2026 Q4–2027 Q1 | v6000:** Torso Humanoid VSA/HDC Platform (With Arms)
- **2027 Q1–Q4 | v7000:** 4-Legged & Bipedal Humanoid Platforms

Phase III: Total Realization (Future Trajectory)

- **2028–2029 | v8000:** Standard Robotics Leading-Edge Platform
- **2030 | v9000:** Enhanced *Bubo Sapiens* Architecture
- **2034 | v10000:** Ultimate Biomimetic (Clone/Kai/Frubber Realization)



The Convergence Locus (2026)

We are currently at the critical inflection point where our proven bare-metal *Bubo* chassis engineering converges with our newly developed 7-tier Neurotronic Brain.

The Neurobotics Plan

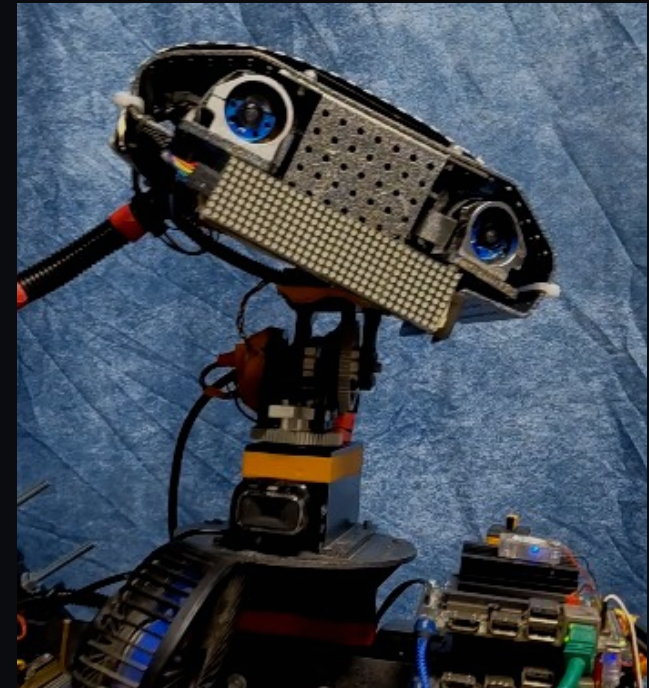


Business Ready LLC

We are a registered LLC with branding, marketing, compliance, and an engineering roadmap built right in. We currently have the working OSS lower tiers, working demo chassis robot, trademark, and LLC registration ready to prove concepts. We are looking to develop the PIC64/Polarfire Neurotronic Brain with a physical target of 100mm by 100 mm and <25W total peak power draw. We operate on a lean, zero-waste budget.

Capital Allocation Breakdown:

- **Core R&D & Hardware Prototyping (50%):**
Development of the PIC64/Polarfire Neurotronic Brain substrate (<25W peak target).
- **IP Protection & Regulatory Compliance (10%):**
Domestic/International patent filing, trademarks, and federal validation.
- **Operational Runway & G&A (40%):** Securing the laboratory workspace, procuring dev tools for PIC64/RISC-V/Polarfire, and standard operational overhead.



Defense-Grade Compliance



Zero-Trust Air-Gapped Enclave

Operating on a "Deny All, Complain Loudly" paradigm, the system features enforced FIPS 140-3 cryptography and SELinux Mandatory Access Controls (MAC). We have embedded the entire DevSecOps pipeline—including Perforce, Gitea, Jenkins, JFrog Artifactory, and Syft SBOM generation containerized via Podman—directly into the bare-metal hardware, creating a hardened, DISA-STIG compliant, defense-in-depth supply chain.

The Invitation

We are what you have been waiting for. Witness a live demonstration of the future of cognitive science, and the next era of biologically-inspired deterministic systems.

Neurobotics - The Future is Here. Real. Now.

Phase I Alpha Demo is officially READY. Available for a short introductory meeting to discuss integration, seed funding, and a physical demo of the prototype chassis.

REAL - DETERMINISTIC - AUTONOMOUS

www.neuroboticssci.ai | www.github.com/neurobotics-sci/neurobotics-biological-substrate

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